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Data Analytics

Data preparation and customer analytics

Conduct analysis on your client's transaction dataset and identify customer purchasing behaviours to generate insights and provide commercial recommendations.

Advanced | 1-2 Hours

Experimentation and uplift testing

Extend your analysis from Task 1 to help you identify benchmark stores that allow you to test the impact of the trial store layouts on customer sales.

Advanced | 1-2 Hours

Analytics and commercial application

Use your analytics and insights from Task 1 and 2 to prepare a report for your client, the Category Manager.

Advanced | 1-2 Hours

Data Preparation And Customer Analytics

Task Overview

What you'll learn

- Understand how to examine and clean transaction and customer data.
- Learn to identify customer segments based on purchasing behavior.
- Gain experience in creating charts and graphs to present data insights.
- Learn how to derive commercial recommendations from data analysis.

What you'll do

- Analyze transaction and customer data to identify trends and inconsistencies.
- Develop metrics and examine sales drivers to gain insights into overall sales performance.
- Create visualizations and prepare findings to formulate a clear recommendation for the client's strategy.

Hear from a team member at Quantium introduce the task

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Advanced 1-2 Hours

Here is the background information on your task

You are part of Quantium's retail analytics team and have been approached by your client, the Category Manager for chips, who wants to better understand the types of customers who purchase chips and their purchasing behaviour within the region.

The insights from your analysis will feed into the supermarket's strategic plan for the chip category in the next half year.

You have received the following email from your manager, Zilinka.

Hi,

Welcome again to the team, we love having new graduates join us!

I just wanted to send a quick follow up from our conversation earlier with a few pointers around the key areas of this task to make sure we set you up for success.

Below I have outlined your main tasks along with what we should be looking for in the data for each.

Examine transaction data – look for inconsistencies, missing data across the data set, outliers, correctly identified category items, numeric data across all tables. If you determine any anomalies make the necessary changes in the dataset and save it. Having clean data will help when it comes to your analysis.

Examine customer data – check for similar issues in the customer data, look for nulls and when you are happy merge the transaction and customer data together so it's ready for the analysis ensuring you save your files along the way.

Data analysis and customer segments – in your analysis make sure you define the metrics – look at total sales, drivers of sales, where the highest sales are coming from etc. Explore the data, create charts and graphs as well as noting any interesting trends and/or insights you find. These will all form part of our report to Julia.

Deep dive into customer segments – define your recommendation from your insights, determine which segments we should be targeting, if packet sizes are relative and form an overall conclusion based on your analysis.

Make sure you save your analysis in the CSV files and your visualisations – we will need them for our report. If you could work on this analysis and send me your initial findings by end of next week that would be great.

Looking forward to reviewing your work.

Thanks

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Data Preparation And Customer Analytics

Here is your task

We need to present a strategic recommendation to Julia that is supported by data which she can then use for the upcoming category review. However, to do so, we need to analyse the data to understand the current purchasing trends and behaviours. The client is particularly interested in customer segments and their chip purchasing behaviour. Consider what metrics would help describe the customers' purchasing behaviour.

We have chosen to complete this task in R, however you will also find Python to be a useful tool in this piece of analytics. If you aren't familiar with R or Python we would recommend searching a few online courses to help get you started. We have also provided an R solution template if you want some assistance in getting through this Task. Whilst its possible to complete the task in Excel you may find the size of the data and the nature of the tasks is such that it is more difficult to complete in Excel.

To get started, download the resource csv data files below and begin performing high-level data checks such as:

- Creating and interpreting high-level summaries of the data
- Finding outliers and removing these (if applicable)
- Checking data formats and correcting (if applicable)

You will also want to derive extra features such as pack size and brand name from the data and define metrics of interest to enable you to draw insights on who spends on chips and what drives spends for each customer segment. Remember, our end goal is to form a strategy based on the findings to provide a clear recommendation to Julia the Category Manager so make sure your insights can have a commercial application.

As we are in the early stages of this analysis Zilinka has asked us to submit our initial findings, so please save your code as a .pdf file and upload it to unlock the model answer.

Note: that this is an open-ended case study that can be approached in many ways. Example answer is in R.

Here are some resources to help you

Here are some resources to help you

Some helpful definitions:

LIFESTAGE: Customer attribute that identifies whether a customer has a family or not and what point in life they are at e.g. are their children in pre-school/primary/secondary school.

PREMIUM_CUSTOMER: Customer segmentation used to differentiate shoppers by the price point of products they buy and the types of products they buy. It is used to identify whether customers may spend more for quality or brand or whether they will purchase the cheapest options.

Pro analytics tip: While the data set would not normally be considered large, some operations may still take some time to run.

This is normal and to be expected when dealing with data sets at Quantium. Whilst your analysis is based on the whole data set you may want to try some strategies to run sample solutions on a smaller subset of data I.e. create a sample solution using some of the data while it uploads.

QVI Transaction Data — Download File

This file is for Forage simulation use only.

CSV QVI Purchase Behaviour — Download File

This file is for Forage simulation use only.

PDF R Solution Template — View PDF

This file is for Forage simulation use only.

Download "R" Software — Visit Website

This link will open in a new tab.

Your Submission

Upload your Code in PDF

Quantium_Analytics.pdf

Allowed file types DOC, DOCX, TXT, ODT, TEX, RTF, WPD, PAGES, PDF, MD, MARKDOWN

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Experimentation And Uplift Testing

1 2 3 4 5

Task Overview

What you'll learn

- Understand experimentation and uplift testing, comparing trial and control stores.
- Learn control store selection based on defined metrics.
- Gain experience in data visualization.
- Perform statistical analysis to assess sales differences and formulate recommendations.

What you'll do

- Define metrics to select control stores.
- Analyze trial stores against controls.
- Use R/Python for data analysis and visualization and summarise findings and provide recommendations.

Hear from a team member at Quantium introduce the task

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Experimentation And Uplift Testing

< 1 2 3 4 5 >

Here is the background information on your task

We suggest you complete Task 1 before proceeding with this task.

You are part of Quantium's retail analytics team and have been approached by your client. The Category Manager for Chips, has asked us to test the impact of the new trial layouts with a data driven recommendation to whether or not the trial layout should be rolled out to all their stores.

You have received the following email from Zilinka.

Hi,

Thanks for your feedback earlier, I'm glad you find my follow up emails helpful in ensuring your on the right track.

For this part of the project we will be examining the performance in trial vs control stores to provide a recommendation for each location based on our insight. Below are some of the areas I want you to focus on, of course if you discover any other interesting insights feel free to include them in your findings.

Select control stores – explore the data and define metrics for your control store selection – think about what would make them a control store. Look at the drivers and make sure you visualise these in a graph to better determine if they are suited. For this piece it may even be worth creating a function to help you.

Assessment of the trial – this one should give you some interesting insights into each of the stores, check each trial store individually in comparison with the control store to get a clear view of its overall performance. We want to know if the trial stores were successful or not.

Collate findings – summarise your findings for each store and provide an recommendation that we can share with Julia outlining the impact on sales during the trial period.

Remember when working with a client visualisations are key to helping them understand the data. Be sure to save all your visualisations so we can use them later in our report. We are presenting to our client in 3 weeks so if you could submit your analysis by mid next week that will give us great amount of time to discuss findings and pull together the report.

Keep up the good work!

Thanks,

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Experimentation And Uplift Testing

Here is your task

Julia has asked us to evaluate the performance of a store trial which was performed in stores 77, 86 and 88.

We have chosen to complete this task in R, however, you will also find Python to be a useful tool in this piece of analytics. We have also provided an R solution template if you want some assistance in getting through this Task.

To get started, use the QVI_data dataset below or your output from task 1 and consider the monthly sales experience of each store.

This can be broken down by:

- total sales revenue
- total number of customers
- average number of transactions per customer

Create a measure to compare different control stores to each of the trial stores to do this write a function to reduce having to re-do the analysis for each trial store. Consider using Pearson correlations or a metric such as a magnitude distance e.g. 1- (Observed distance – minimum distance)/(Maximum distance – minimum distance) as a measure.

Once you have selected your control stores, compare each trial and control pair during the trial period. You want to test if total sales are significantly different in the trial period and if so, check if the driver of change is more purchasing customers or more purchases per customers etc.

As we are in the early stages of this analysis Zilinka has asked us to submit our initial findings, so please save your code as a .pdf file and upload it to unlock the example answer.

Here are some resources to help you

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Pro analytics Tip: While the data set would not normally be considered large some operations may still take some time to run.

This is normal and to be expected when dealing with data sets at Quantium. Whilst your analysis is based on the whole data set you may want to try some strategies to try running sample solutions on a smaller subset of data I.e. create a sample solution using some of the data while it uploads.

We think you've got everything you need for this task but if you're stuck and want a hand, drop us an email at admin@insidesherpa.com mentioning this task.

PDF R Solution Template View PDF

This file is for Forage simulation use only.

CSV QVI_data Download File

Download the data file

Your Submission

Upload your Code in PDF

Task_2_Experimentation_and_Uplift_Testing.pdf

Allowed file types DOC, DOCX, TXT, ODT, TEX, RTF, WPD, PAGES, PDF, MD, MARKDOWN

Submission complete, great work!

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Analytics And Commercial Application

1 2 3 ... 6

Task Overview

What you'll learn

- Gain experience in preparing a client report based on data analysis.
- Understand the "Pyramid Principles" framework for report structure.
- Practice creating data visualizations and key callouts.
- Develop skills in translating data into actionable insights and recommendations.

What you'll do

- Use the Pyramid Principles framework for structuring the report.
- Submit a report that incorporates data visualizations, key insights, and recommendations.

Hear from a team member at Quantium introduce the task

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Analytics And Commercial Application

< 1 2 3 ... 6 >

Here is the background information on your task

We suggest you complete Tasks 1 and 2 before proceeding with this task.

Task 3 is targeted specifically at building your ability to recognise commercial, actionable insights from your analysis and displaying it in a clear and concise way for your client, with minimal jargon. At Quantium, our analyst graduates sometimes work as what we like to call “hybrids” (a mix of analyst and consultant duties) so developing your presentation skills early is a huge win!

As both technical tasks 1 and 2 were open ended in terms of insights, this model answer will focus on the layout and the order of your inclusions, including where to include graphs, taglines, written insights and recommendations.

As part of Quantium’s retail analytics team, you have been conducting a range of analysis on transaction and purchase behaviour data to provide key recommendations to your client, the Category Manager of chips, who is putting together their strategic plan.

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Advanced 1-2 Hours

Analytics And Commercial Application

‹ 1 2 3 4 5 6 ›

Here is your task

With our project coming to an end its time for us to send a report to Julia, based on our analytics from the previous tasks. We want to provide her with insights and recommendations that she can use when developing the strategic plan for the next half year.

As best practice at Quantium, we like to use the "Pyramid Principles" framework when putting together a report for our clients. If you are not already familiar with this framework you can find quick introductions on by searching form them on the internet.

For this report, we need to include data visualisations, key callouts, insights as well as recommendations and/or next steps.

We recommend you use a tool like PowerPoint (or similar) to create your report, we have also provided you a template to help get you started.

One way to create your visualisations (graphs, charts etc) is to save your output files from task 1 & 2 and create generate your charts in Excel. You may find it helpful to create chart templates however that is not essential. We are looking for consistency here – if a series is blue in one chart it needs to remain blue throughout the presentation. Once you have created the charts in your Excel file, paste them into your PowerPoint in the order that suits best.

When you have finished creating your report, remember to save it as a PDF to submit in the next page. **Below, use the text input to write your cover email to the client you will send with the report.**

Here are some resources to help you

Here are some useful resources on the "Pyramid Principles" framework:

We have also attached a PowerPoint template that you might like to use to complete the task. Keep in mind the key considerations for a presentation:

- Data literacy level of your audience
- Table of contents / agenda

considerations for a presentation:

- Data literacy level of your audience
- Table of contents / agenda
- Problem statement / purpose
- Overview and context
- Content balance
- Layout and content display
- Summary / next steps

The Pyramid Principle Visit Website

This link will open in a new tab.

A Real-Life Example of The Pyramid Principle In PowerPoint Visit Website

This link will open in a new tab.

PowerPoint Template Download File

Please feel free to use this template as a guide in preparing your presentation

Your response

Subject: Submission – Quantium Data Analytics Virtual Experience (Task 3)

Dear Julia,

Please find attached my presentation for **Task 3: Analytics and Commercial Application** of the Quantium Data Analytics Virtual Experience.

This report summarizes the key findings from my analysis, including customer insights, brand performance, trial store results, and recommendations for the next half-year strategy. The presentation has been prepared based on the analyses completed in Tasks 1 and 2.

Thank you for reviewing my submission. I appreciate the opportunity to complete this virtual experience.

Kind regards,

Kiran Kalisetti